A Mini Project Report

on

CampusHubX

Submitted in partial fulfilment of the requirement of

Bachelor of Technology (S.Y)

in

COMPUTER ENGINEERING

submitted by

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2025-2026

Approval Sheet

This is to certify that the Mini Project entitled

CampusHubX

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Declaration

We declare that this written submission for S.Y. Mini Project entitled **"CampusHubX"** represent our ideas in our own words and where others' ideas or words have been included. We have adequately cited and referenced the original sources. We also declared that we have adhere to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any ideas / data / fact / source in our submission. We understand that any violation of the above will cause for disciplinary action by institute and evoke penal action from the sources which have thus not been properly cited or from whom paper permission have not been taken when needed.

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ABSTRACT

Campus life presents students with a variety of daily challenges, including navigating in campus, accessing academic resources, having official platform for community discussion and managing canteen services. These fragmented systems often lead to inefficiencies, increased time consumption, and a lack of seamless access to essential services.To address these issues, CampusHubX is introduced as an integrated and intelligent digital platform that consolidates resources sharing and discussion, digital canteen coupon and digital navigation system into a unified, user-friendly ecosystem. Developed using robust backend technologies such as PHP and supported by a centralized MySQL database, the system ensures efficient data processing, real-time updates, and secure transactions.By centralizing key campus functionalities, CampusHubX significantly reduces operational complexity, enhances user convenience, and fosters collaboration among users. The platform also promotes sustainability by encouraging the reuse of academic resources and minimizing reliance on paper-based processes. Ultimately, CampusHubX contributes to the digital transformation of educational institutions by streamlining campus operations, improving accessibility, and delivering a smarter, more connected campus experience.

Keywords: CampusHubX, Canteen Coupon, Resource Sharing and Discussion, Digital Navigation, Web Technologies,Dijkstra Algorithm, Sustainable Systems, User Convenience

Contents

Chapter 1

Introduction

1.1 Background

Not just students but also staff or any other people in college often struggle with various problems like canteen queue, validated resources, and finding a classroom in their college life, which leads to underutilization of their time and effort. As a result, the effort and time they could utilize for work are being wasted in this struggle. Therefore, building a platform that will address these problems by integrating the services at one place. It provides users with access to the resources, including a discussion platform, ordering canteen food with a token system also navigating through campus using the Dijkstra algorithm.

1.2 Motivation

The main motivation is to improve the lives of campus users by saving time and bringing the necessary facilities to an integrated platform. Including providing quick access to campus locations using the Dijkstra algorithm. In addition to encouraging resources sharing and building a community to discuss various topics or solve queries.Reducing the long queues creating chaos in the canteen by building a safe transaction-based Canteen Coupon generation.Overall, to enhance the student experience with a single connected system.

1.3 Aim and Objective

• Aim:

To develop an integrated web platform that simplifies and enhances campus life by providing Dijskastra-based navigation, resource sharing and discusion and a digital canteen coupon system.

• Objective

1. To design an interactive digital navigation system.
2. To build a community for resource sharing and discussion
3. To implement an OR-based canteen coupon system.

1.4 Report Outline

The following outline provides an overview of the chapters and sections included in this report, illustrating the scope and features of the developed

1. Abstract: Provides a brief summary of the project, its objectives, and outcomes.
2. Introduction: Introduces the project concept, background, and overall purpose.
3. Literature Survey: Reviews existing research and related systems relevant to the project.
4. Existing System: Describes current solutions and their limitations.
5. Problem Definition: Identifies the key challenges and gaps addressed by the project.
6. Proposed System: Explains the new system's features, functionality, and innovations.
7. System Architecture: Illustrates the structural design and workflow of the developed system.
8. References: Lists the sources, research papers, and materials referred to in the report.

Chapter 2

Study of The System

2.1 Literature Survey

2.1.1 Localized Navigation System for college campus

The research paper[1] focuses on creating a localized navigation system for college campuses. Its main goal is to help students, staff, and visitors navigate indoor campus spaces efficiently using modern tracking technologies. The paper reviews existing methods, compares their accuracy, cost, infrastructure requirements, and highlights the challenges of implementing indoor navigation systems in real-world campus environments.[1[]]

2.1.2 Design and Development of a Digital Platform for Peer-to-Peer Book Exchange

The research paper[2] presents the design and development of a digital platform (app) that enables peer-to-peer book sharing. Users can list their books, search for books, request to borrow, and rate both books and other users, aiming to make book exchange more convenient, cost-effective, and trustworthy.[2]

2.1.3 Smart Canteen Management System

The Research paper[3] proposes a Smart Canteen Management System where RFID technology is used to automate canteen operations. The system enables user identification, automated billing, and access control, aiming to improve efficiency and convenience for both staff and users.[3]

2.1.4 Integrated Campus Management System using Mobile Application

The paper[4] presents a campus navigation system integrated with AI-based food services. It aims to help students locate empty seats in cafeterias and recommend food items based on preferences and feedback. The system also analyzes customer comments to rate food quality.[4]

2.2 Existing System

2.2.1 Smart Canteen Management System

The Smart Canteen Management System is designed to automate food ordering, billing, and user authentication processes within a college campus. It typically uses RFID key cards or smart ID cards to identify users, allowing them to make cashless transactions. The system records every purchase in a centralized database for analysis and transparency.

2.2.2 Digital Platform for Peer-to-Peer Book Exchange

This type of platform allows students to list, search, and request books from peers through a mobile app or web interface. Users can rate each other and leave feedback after a successful exchange, improving trust and transparency. Such systems promote cost-effective learning by reducing the need to purchase new books.

2.2.3 Localized Navigation System for College Campus – AI Food Recommendation

This prototype uses AI and Machine Learning algorithms such as TensorFlow for empty seat detection, Collaborative Filtering for food recommendations, and Logistic Regression for feedback analysis. It assists students in finding available seating areas, recommending food options, and collecting user feedback for improvement.

2.3 Table of Comparison

Sr. No.	Research Paper	Objectives	Methodology	Gaps
1	Smart Canteen Management System	To automate food ordering, billing, and user authentication in college canteens using RFID or smart cards for cashless transactions.	Uses RFID technology, database management systems (MySQL/SQLite), and web or mobile interfaces for order placement and data storage. Data analytics used for purchase tracking and transparency.	Uses RFID technology, database management systems (MySQL/SQLite), and web or mobile interfaces for order placement and data storage. Data analytics used for purchase tracking and transparency.
2	Digital Platform for Peer-to-Peer Book Exchange	To enable students to share, sell, or borrow academic books among peers through an online platform, reducing learning costs.	Implemented using mobile/web development frameworks (like Android Studio or React), database backend (Firebase/MySQL), and rating-feedback algorithms for trust and transparency.	Implemented using mobile/web development frameworks (like Android Studio or React), database backend (Firebase/MySQL), and rating-feedback algorithms for trust and transparency.
3	Localized Navigation System for College Campus – AI Food Recommendation	To help students find available seating areas and recommend food options using AI and ML algorithms.	Uses TensorFlow for empty seat detection, Collaborative Filtering for food recommendation, and Logistic Regression for feedback analysis. Integrates real-time data processing and AI-based decision-making.	Uses TensorFlow for empty seat detection, Collaborative Filtering for food recommendation, and Logistic Regression for feedback analysis. Integrates real-time data processing and AI-based decision-making.

Figure 2.1: Comparison of Research Papers

Chapter 3

Proposed System

3.1 Problem Statement

In most college campuses, users face difficulty accessing essential campus facilities. They often struggle to locate classrooms or departments, finding relevant resources and solving query facilities, waiting for long queues in canteen. Currently, these services are handled manually or through separate systems, leading to inconvenience, time wastage, and communication gaps. CampusHubX aims to solve these problems by integrating multiple features, such as the Digital Navigation System, Resource Sharing, Discussion, and the Digital Coupon System, into a single unified platform. This system provides path-based campus navigation, simplifies online resource sharing among students, and offers a QR-based canteen coupon generation process to make campus life smoother and more organized.

3.2 Scope

Developing a web-based platform accessible to all students and staff within the campus network.
Providing Smart Navigation that helps students locate classrooms, labs, offices, and other facilities within the campus using interactive maps.
Building a Book Sharing Community that enables students to lend, borrow, or request academic books from peers.
Creating a Digital Coupon System for canteen management where stu-

dents can purchase, redeem, and track coupons electronically.
Ensuring secure user authentication for both students and canteen staff.
Offering an admin panel for managing users, books, maps, and coupon records.

3.3 Hardware Requirements

Processor: Intel i3 or higher
RAM: Minimum 4 GB
Storage: 100 GB Hard Disk or SSD
Network: Wi-Fi for accessing the web application
Server: Local Host server (XAMPP)
Input Devices: Keyboard and Mouse
Output Devices: Monitor or Mobile Screen

3.4 Software Requirements

Operating System: Windows 10/11, Linux, or macOS
Frontend: HTML, CSS, JavaScript
Backend: PHP
Database: MySQL
Web Server: Apache
IDE / Editor: Visual Studio Code
Version Control: Git and GitHub
Browser: Google Chrome

Chapter 4

Design of the System

4.1 Approach

The proposed system adopts a modular design approach to improve campus management and enhance user convenience. It is structured around three primary modules: Digital Navigation System, Resource Sharing and Discussion, and Digital Canteen Coupon. The Digital Navigation module generates a path using the Dijkstra algorithm and helps users in locating various classrooms. It provides optimal routes for efficient movement across the campus. The Resource Sharing and Discussion module enables users to search, view, upload, and download resources within the campus community, promoting collaboration and resource reuse. With that, it also has functionality to post queries and join the conversation. The Digital Canteen Coupon module allows users to explore menu options, select food items, add the items to the cart, and generate digital coupons, which ensures quick and hassle-free transactions. All modules are interconnected through a centralized database, which ensures seamless data integration, consistency, and easy accessibility across the system. User requests are processed efficiently, and the corresponding results are delivered through an intuitive interface, completing the workflow smoothly. Overall, this modular approach ensures flexibility, scalability, and efficient system performance, making CampusHubX a reliable solution for simplifying campus activities.

4.1.1 Flow Chart

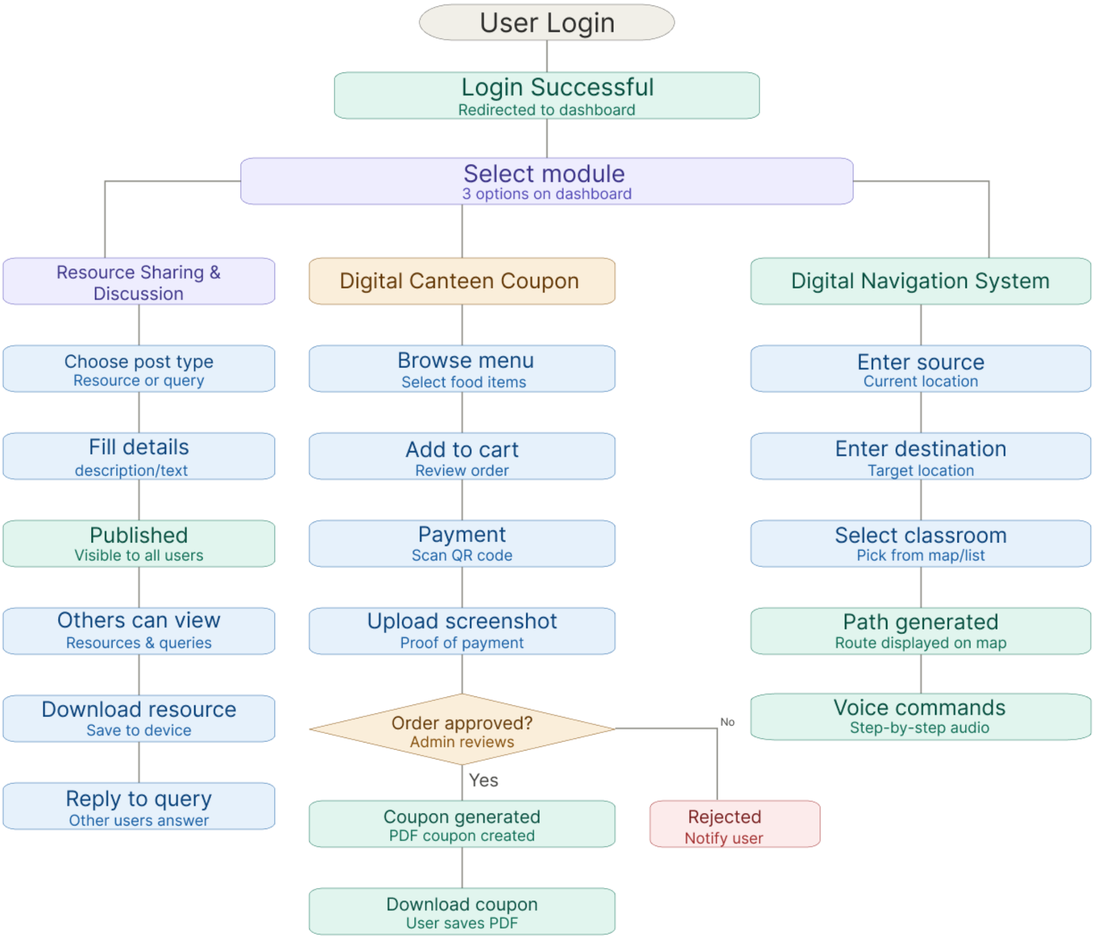

Figure 4.1: CampusHubX Flowchart

4.2 Technology Stack

4.2.1 Tools Required

Front End

1. HTML – For creating the basic structure of the web pages.
2. CSS – For styling and improving the user interface design.
3. JavaScript – For adding interactivity and client-side validation.
4. Bootstrap - For designing responsive layouts and components like navigation bars, forms, and buttons.

Back End

1. MySQL – Used to store and manage all application data in a structured format.
2. PHP – Used to process data, connect with the database, and control the working of the website.

4.3 System

The CampusHubX system architecture provides an integrated framework connecting multiple campus services such as Resource Sharing and Discussion, Digital Canteen Coupons, and Campus Navigation through a unified web-based system. The process begins with users, including students, faculty, admin, and canteen admin, who can log in, register, manage profiles, access resources, generate coupons, and navigate across campus. These user actions are handled through the frontend web application built using HTML, CSS, JavaScript, and Bootstrap, which provides an interactive interface, collects input, performs basic validation, and sends requests to the backend. These requests are routed to the backend system developed using PHP and AJAX, which manages authentication, enforces role-based access control, executes core business logic, and ensures proper communication between modules and the database. The backend processes each request, validates user roles, and directs it to the appropriate module, such as Resource Sharing and Discussion, Canteen Coupons, or Campus Navigation. It also ensures secure and efficient data exchange between the frontend and the database. The database layer, implemented using MySQL, stores all essential data, including user credentials, profile information, uploaded resources, categories, transaction records, and order details. It performs CRUD operations, maintains data consistency, and ensures integrity for reliable and fast data retrieval.

Each core module interacts with the backend through structured request handling. The Resource Sharing and Discussion module enables users to upload, categorize, and download resources while sending notifications. The Canteen Coupons module allows users to view menus, generate QR-based coupons, process payments, and track orders. The campus navigation module fetches user location, calculates optimal paths,

and displays routes using the Dijkstra algorithm.
Overall, this modular architecture ensures efficient data flow, scalability, secure access, and smooth user interaction, making CampusHubX a unified, reliable, and effective solution for campus management.

4.3.1 Block Diagram

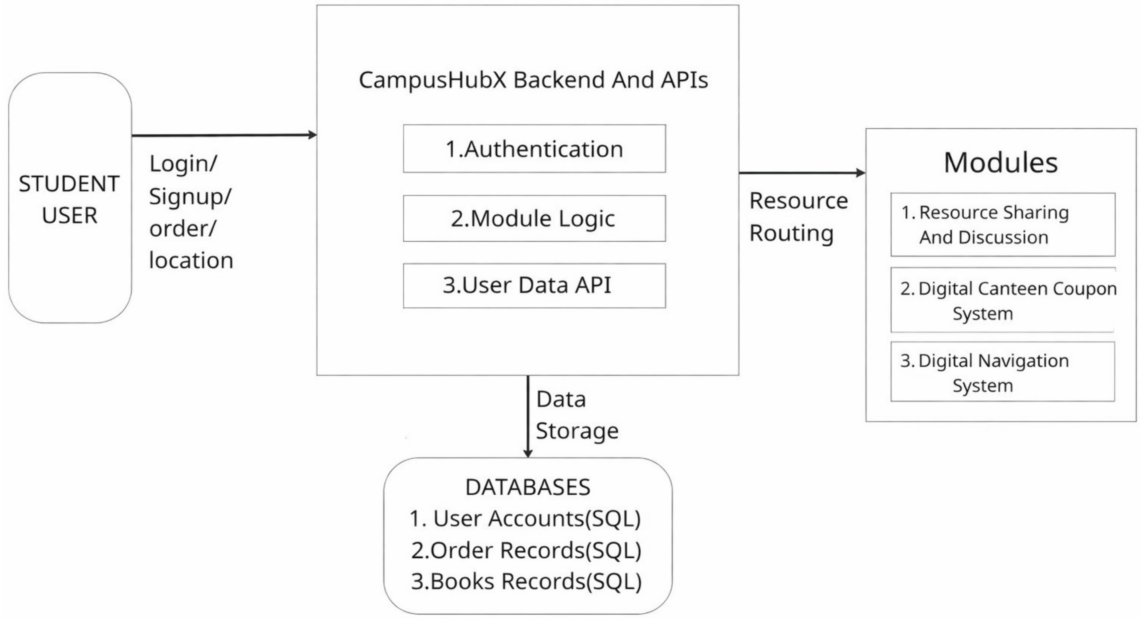

Figure 4.2: CampusHubX Block Diagram

The User interacts with the system by performing actions like login, signup, placing orders, navigating within the campus, and uploading and accessing shared resources. These requests are sent to the Backend and APIs, which handles:
Authentication – verifies user identity
Module Logic – processes different features
User Data API – manages data communication
The backend then routes requests to different Modules, such as: Resource Sharing and Discussion, Digital Canteen Coupon System, Digital Navigation System. All data is stored in the Database, including User accounts, Order History, Resource/Discussion records.

Chapter 5

Conclusion and Future Scope

CampusHubX integrates essential campus services into a single, user-friendly digital platform, combining Resource Sharing and Discussion, Digital Canteen Coupon and Digital Navigation System. It simplifies daily student activities by reducing manual processes and saving time of the users. The web-based architecture ensures accessibility, scalability, and smooth user interaction. Additionally, the platform supports sustainability by encouraging paperless operations and reuse of study materials. The system follows a modular design where all the 3 modules work seamlessly through a centralized backend, ensuring efficient data handling and reliable performance, thereby improving overall campus efficiency and user convenience. .

In the future, CampusHubX can evolve into a comprehensive Smart Campus Management System by incorporating features such as event notifications, attendance monitoring,etc. The Canteen Coupon generation can be done through payment gateway. The smart navigation module can be upgraded with technologies like GPS integration or augmented reality (AR) based indoor positioning for more accurate location guidance. With these advancements, CampusHubX can become a fully integrated, intelligent platform that enhances campus infrastructure, user experience, and overall connectivity.

Acknowledgement

Success of a project like this involving high technical expertise, patience, and massive support of guides, is possible when team members work to-gather. We take this opportunity to express our gratitude to those who have been instrumental in the successful completion of this project. We would like to appreciate the constant interest and support of our mentor guide **Meenal Bhoyar** in our project and aiding us in developing a flair for the field of Web Development. We would always cherish the journey of transforming the idea of our project into reality. We would like to show our appreciation to **Meenal Bhoyar** for his/her tremendous support and help, without whom this project would have reached nowhere. We would also like to thank our project coordinator **Mrs.Chetana Badgujar** for providing us with regular inputs about documentation and project timeline. A big thanks to our **HOD Dr. M. Kiruthika** for all the encouragement given to our team. We would also like to thank our principal, **Dr. S. M. Khot**, and our college, **Fr.C.Rodrigues Institute of Technology, Vashi**, for giving us the opportunity and the environment to learn and grow.

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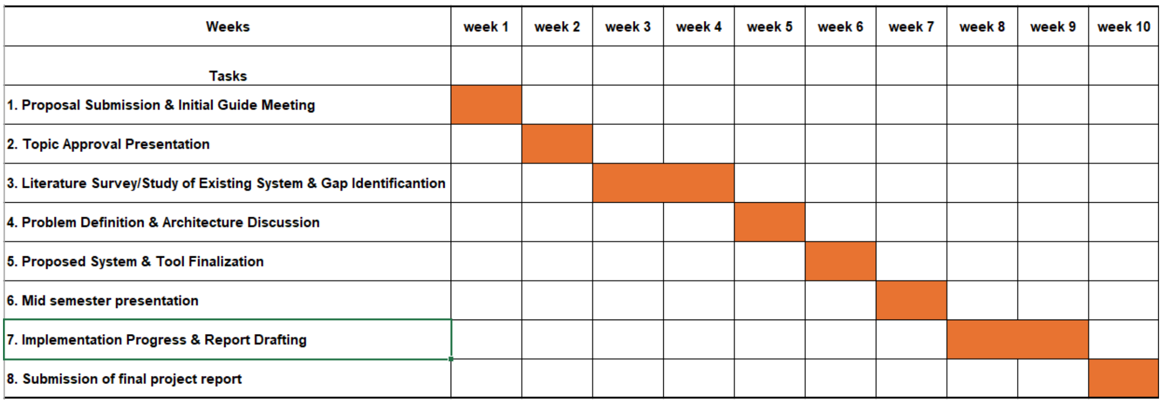
.1 Appendix A: Timeline Chart



Weeks	week 1	week 2	week 3	week 4	week 5	week 6	week 7	week 8	week 9	week 10
Tasks										
1. Proposal Submission & Initial Guide Meeting										
2. Topic Approval Presentation										
3. Literature Survey/Study of Existing System & Gap Identificantion										
4. Problem Definition & Architecture Discussion										
5. Proposed System & Tool Finalization										
6. Mid semester presentation										
7. Implementation Progress & Report Drafting										
8. Submission of final project report										

Figure 1: Timeline Chart